

Experiment 4: Interrupt Control Using the PIC32MX370F512L Microcontroller

Objective: Use the interrupt capabilities of the microcontroller to create a precise time delay.

Tasks: Use the given project template in order to accomplish the following tasks.

1. Run the project template program on the PIC32 board and observe that **LED0** turns on. Without making any changes, review the program code and note that this LED is actually flashing at a very high frequency.
2. Configure the Timer 1 control registers in order to enable use of the timer. This entails initializing registers **T1CON** and **PR1** to the appropriate bit values. Choose the 1:256 prescaler and a timer period of 0.5 sec.
3. Configure the Timer 1 interrupt control registers in order to trigger and service interrupts. This entails initializing registers **IEC0** and **IPC1** to the appropriate bit values. Set the timer interrupt priority level to 2. Note that the code for this step, as well as the previous one, should be written in the Initial Subroutine in your program.
4. Identify the section of code containing the Timer 1 Interrupt Service Routine (ISR) located at the bottom of the program. This ISR is responsible for acknowledging and servicing the interrupt from the timer rollover. Inside this ISR, write the code to clear the Timer 1 interrupt flag bit in the **IFS0** register. This will ensure that the interrupt can be triggered repeatedly.
5. If all the previous steps have been coded correctly up to this point, then **LED1** should be flashing every 0.5 sec. At this point, the Timer 1 interrupt is set up and ready to be used to create a delay.
6. Inside the main loop in the program, write the code for a loop that breaks out every time the interrupt trigger occurs, which signals that the timer has rolled over. This loop should be created using conditional statements in order to loop indefinitely until the interrupt is triggered by Timer 1. If this is done correctly, **LED0** and **LED1** will blink every 0.5 sec and there will be a precise pulse train, with a 50% duty cycle and a 1 sec period, being generated at pin **JA7** of **PMDA**. Use the oscilloscope to measure this pulse train and show your work to your instructor.
7. Set the timer period to the maximum value by changing the appropriate bits in register **PR1** and run your code again. Use the oscilloscope to measure the pulse train to verify that his maximum period is achieved.

Guidelines:

1. For details about the Timer 1 control registers, consult the datasheet starting in p. 169, and pay close attention to Table 13-1 in pp. 170-172.
2. For details about the Timer 1 interrupt control registers, consult the datasheet starting in p. 65, and pay close attention to the information for **T1- Timer 1** in the Interrupt Source column of Table 7-1 in p. 66.